JUAN C. SANCHEZ-ARIAS, MD, PHD

Data Scientist • Physician-Scientist • HealthTech & AI Systems Leader

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PROFESSIONAL PROFILE

Physician-Scientist, Data Scientist, and Google-Certified Generative AI Leader with over 10 years of combined clinical, biomedical research, and healthtech industry experience. Specialized in engineering production-grade Generative AI pipelines, RAG architectures, and biomedical knowledge graphs that accelerate preclinical drug discovery and solve complex biological data problems. Proven track record leading cross-functional teams of PhD scientists and ML engineers, bridging clinical insights with modern data engineering to build compliant, scalable, and impactful AI products.

CORE COMPETENCIES & TECHNICAL SKILLS

AI & LLM Systems: Generative AI, Agentic AI & MCP (Model Context Protocol), Prompt Engineering, RAG Frameworks, LLM Evaluation & Benchmarking, Knowledge Graphs, Biomedical NLP.

Programming & Data: Python (pandas, NumPy, scikit-learn, PyTorch/HuggingFace), R (tidyverse, Bioconductor: DESeq2, limma, oligo), SQL (PostgreSQL, BigQuery), REST APIs, Bash, Git.

Life Sciences Domain: Preclinical Drug Discovery, Clinical Medicine, Target Validation, Bio-Ontologies, Live-Cell Bio-Image Processing (ImageJ/Fiji), Computational Biology.

Product & Leadership: Technical Product Ownership, Scientific Publisher Compliance & Data Governance, Cross-Functional Team Leadership, Agile/Scrum, Stakeholder Management, GCP, Docker.

WORK EXPERIENCE

Data Scientist — BenchSci

Jun 2026 – Present | Remote (Toronto, ON)

Enterprise AI platform accelerating preclinical drug discovery for leading global pharmaceutical organizations.

- Architect production-grade Generative AI, RAG, and agentic workflows to extract, synthesize, and contextualize high-dimensional preclinical and therapeutics data for top global pharmaceutical clients.
- Expand BenchSci's product ecosystem via programmatic access and agentic integrations, contributing to **Model Context Protocol (MCP)** tooling and high-throughput API services for preclinical discovery workflows.
- Lead technical assessments and governance audits across BenchSci data services and pipelines to ensure strict compliance with **scientific publisher requirements**, licensing agreements, and content usage standards.
- Develop automated LLM benchmarking and evaluation pipelines to ensure high scientific accuracy and hallucination mitigation, scaling biomedical knowledge graph coverage across millions of research papers, patents, and internal experimental records.

Science Tech Lead & Team Lead — BenchSci

Jun 2023 – May 2026 | Remote (Toronto, ON)

- Managed and mentored a cross-functional team of PhD scientists and data specialists responsible for ingesting, validating, and structuring proprietary enterprise pharmaceutical datasets.
- Served as **Product Owner** for the Customer Internal Data E2E processing pipeline, dramatically accelerating ingestion velocity and client time-to-value.
- Led the R&D and deployment of Generative AI applications to extract, summarize, and structure unstructured experimental records in Electronic Lab Notebooks (ELNs) and IP documents.
- Acted as subject matter expert for Neuroscience and Oncology therapeutic areas, shaping platform roadmap with executive stakeholders and technical product managers.

Scientist II — BenchSci

Jun 2022 – May 2023 | Remote (Toronto, ON)

- Engineered automated data ingestion and ontology mapping pipelines in Python/SQL for preclinical assays and biological reagents, reducing manual review cycles by over 30%.
- Partnered directly with pharmaceutical partners to map bespoke scientific schemas to standardized knowledge graph ontologies.

WORK EXPERIENCE (CONTINUED)

Postdoctoral Research Fellow (Health Research BC Fellow) — University of Victoria Nov 2020 – Apr 2022 | Victoria, BC

Division of Medical Sciences

- Recipient of the **Michael Smith Health Research BC Trainee Award** investigating cardiac arrhythmia and neurodevelopmental channelopathies (*ANK2* p.S646F) in BC Indigenous communities.
- Performed bulk RNA-seq and microarray differential gene expression analysis in **R / Bioconductor** (DESeq2, limma, oligo), alongside protein-protein interaction (PPI) network and pathway enrichment analyses (R, Cytoscape) to uncover mechanisms of cardiomyocyte calcium homeostasis.
- Executed immunohistochemistry and confocal fluorescence microscopy of embryonic and adult mouse heart sections, bridging preclinical genetics with clinical diagnostic insights.

Graduate Research Assistant (Ph.D. Candidate) — University of Victoria Sep 2015 – Aug 2020 | Victoria, BC

- Built custom Python and ImageJ bio-image analysis pipelines for automated quantification of 3D dendritic spine morphology and synaptic plasticity in cortical neurons.
- Taught neuroanatomy and histology laboratory curricula to 1st/2nd year medical students (Island Medical Program, UBC/UVic).

Pre-Diploma Medical Intern & Clinical Researcher — Hospital Universitario del Valle 2013 – 2014 | Cali, Colombia

- Completed full-time clinical rotations across Internal Medicine, General Surgery, Pediatrics, OB/GYN, and Emergency Care in a tertiary-care university hospital.
- Conducted clinical research and bedside patient evaluations with the Department of Neurosurgery, contributing to clinical protocols and neurocritical care management.

EDUCATION

Doctor of Philosophy (Ph.D.) in Neuroscience — University of Victoria, BC, Canada 2015 – 2020

Doctor of Medicine (MD / Médico y Cirujano) — Universidad del Valle, Cali, Colombia 2007 – 2014

CERTIFICATIONS & PROFESSIONAL TRAINING

- **Basic Life Support (BLS) - HCP, Airway Management, and Oxygen Therapy** — *Canadian Red Cross* Oct 2026
- **Google Data Analytics Professional Specialization** — *Google Cloud / Coursera* 2026
- **Certified Google Generative AI Leader** — *Google Cloud* 2025
- **Genetic Epidemiology & Epigenetics** — *Universidad del Valle* 2022
- **Code in Place (Python Engineering)** — *Stanford University* 2021

SELECTED AWARDS & HONORS

Research Trainee Award — *Michael Smith Health Research BC* 2021 – 2022

Nb Gulila Star Award — *International Gap Junction Conference* 2019

John & Myrtle Tilley Graduate Scholarship — *University of Victoria* 2019

James & Laurette Agnew Memorial Scholarship — *University of Victoria* 2015 – 2019

School of Medicine Yearly Dean's List — *Universidad del Valle* 2007, 2009, 2013, 2014